Dorm Curfew Tracker: A Python-Based Dormitory Monitoring and Notification System

I. Problem Statement / Rationale

The dormitory management system often faces challenges towards efficient student compliance. Traditional methods such as manual logbooks often lack transparency and are prone to human errors. These problems increase the difficulty of improving dormitory tracking systems.

Additionally, this method creates a communication gap between the dormitory managers and guardians, in times of violations or regarding student behavior. And this limitation reduces student accountability and can put students' safety at risk.

To address these concerns, there is a need to have an automated dormitory management system that is reliable and is capable of recording and monitoring real-time dormer activities. This solution aims to improve time accuracy and reduce communication gaps between the people in charge.

II. Project Overview & Relevance

Our project's use is to monitor and record data of students' check-in time while also notifying the dorm manager and the students themselves of their current status eliminating the previously addressed issues. The project single-handedly avoids human errors and replaces traditional logbooks.

Python is our programming language of choice due to its simplicity in all aspects as well as versatility. The project utilizes Python's library zoneinfo allowing datetime to track in real-time, have a set curfew for the program, as well as allow improved readability. Other libraries for example, os were used for atomic renaming and json to have a local database. In the program, hashlib was used for passkeys, specifically SHA-256 algorithm while in order to handle .py files in other folders, pathlib was used to do so.

Furthermore, specific student IDs and passkeys are given to ensure that there is no repetition of data. Lists are given names corresponding to their purpose such as "hist" meaning history and "vio" meaning violation allowing for better data analysis. This follows the PEP 8 - Style Guide for Python reducing the effort of other programmers years from now in their search for the meaning of the list names when accessing our repository on GitHub.

III. Objectives

This project aims to achieve the following objectives:

- To make a system capable of monitoring student check-in and check-out activities.

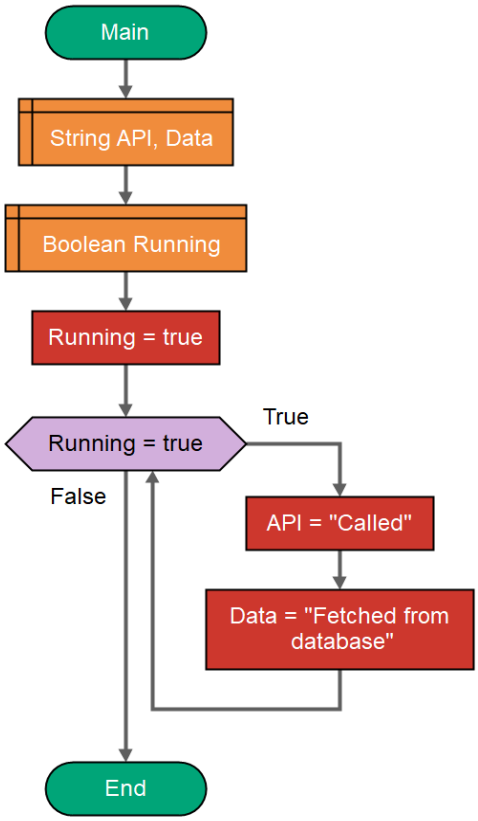
- To have a login feature that differentiates a student from a dorm manager, allowing different access features.
- To create a system that is able to record violations and alert the users and the dorm managers.
- To design a dashboard that can store student information (personal and dormitory details)
- To make a system that can notify guardians in case of dormitory violations.

IV. Methodology / Technical Approach

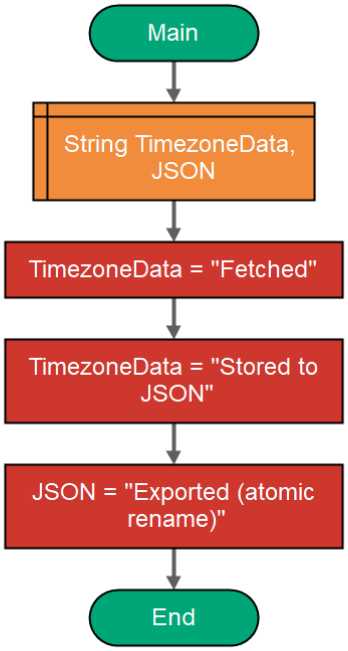
- Time API (Python)
  - Allows the Curfew Tracker to access a timezone API and track the current time when run.
- Main (Python)
  - Allows the user to access the program either as a student or a manager. Requires the ID and pin to access.
- Dorm Manager Management System (Python)
  - Allows the dorm manager to add and remove managers and students, access student information, and student history and violations.
- Student Management System (Python)
  - Allows the intern to check in, out, view their profile, and check their violations.
- Database (Python)
  - Accesses the student check-in history, manager information, student information, and violations.
- History (JSON)
  - Displays the student id, check-in status, and the violations of each student against the curfew policy. Displays the time of the statuses of the students checked by the manager.
- Violations (JSON)
  - Displays the type of violation the student committed against the dorm curfew policy.
- Managers (JSON)
  - Displays the manager name and password hash.
- Students
  - Displays the student's name, room number, password, and check-in status.
- Student Check-In History (JSON)
  - Checks and displays the student's status whether or not they are checked in, if they violated the curfew policy, as well as the time in which the statuses of the students were checked.

V. Flowchart

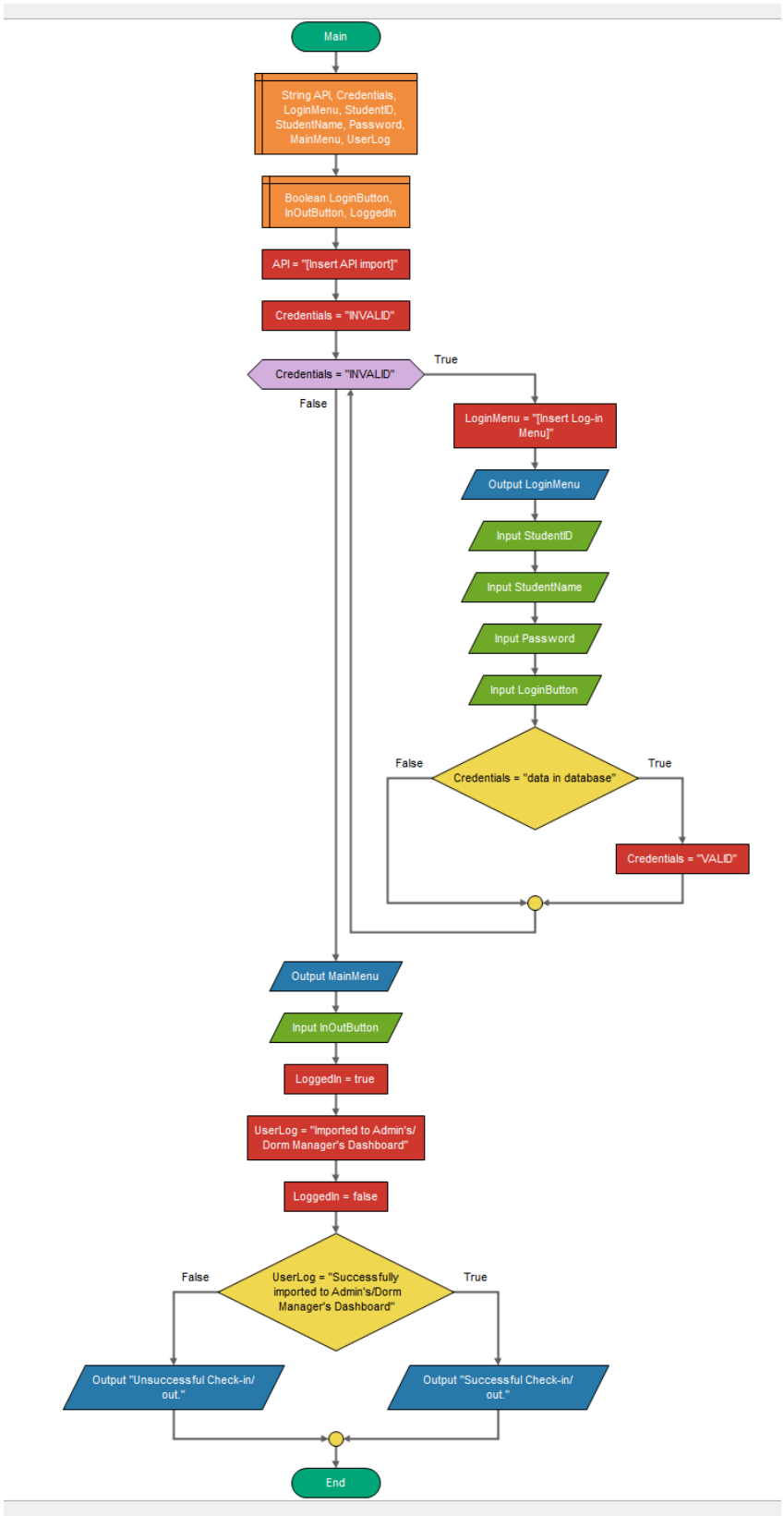
Caller API:



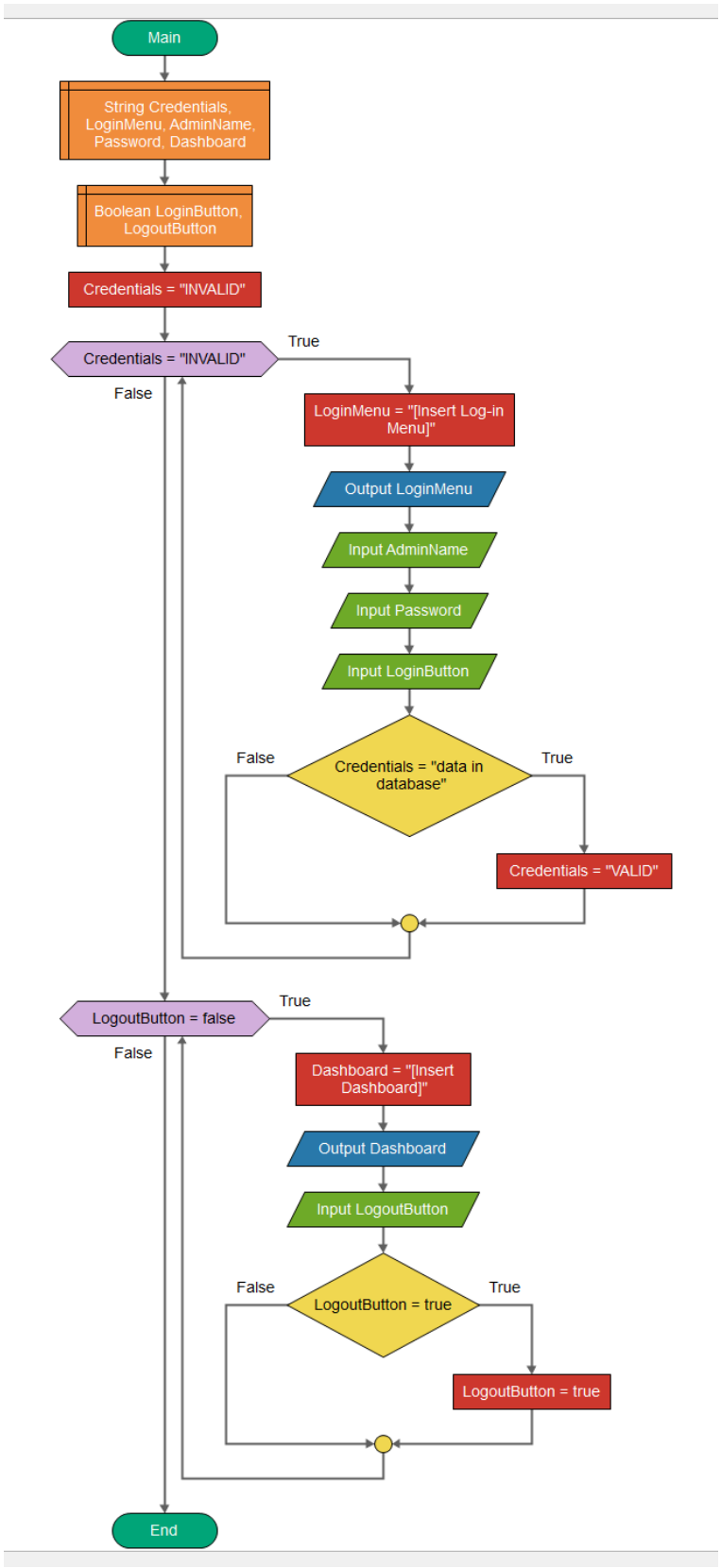
API Function:



Main Function (Student):



Main Function (Admin/Dorm Manager):



VI. Program

```
------INTERFACE------

                [1.] Student
                [2.] Manager
                [3.] Exit

1
Enter your student ID: 14-2024-01
Enter your student pin: 12345

----- Student Dashboard -----
1 : Check-in
2 : Check-out
3 : View Student Profile
4 : View Student Violations
5 : Logout

Choice: 2
Check-out time: 2026-04-19 20:45

----- Student Dashboard -----
1 : Check-in
2 : Check-out
3 : View Student Profile
4 : View Student Violations
5 : Logout
```

```
----- Student Dashboard -----
1 : Check-in
2 : Check-out
3 : View Student Profile
4 : View Student Violations
5 : Logout

Choice: 4
----- Your Violations -----
Student ID: 14-2024-01
Violation Type: Late Check-in
Formatted Time: 2026-04-19 20:17:33
Note: Created by SYSTEM

Student ID: 14-2024-01
Violation Type: Late Check-in
Formatted Time: 2026-04-19 20:20:27
Note: Created by SYSTEM
```

```
----- Student Dashboard -----
1 : Check-in
2 : Check-out
3 : View Student Profile
4 : View Student Violations
5 : Logout

Choice: 4
----- Your Violations -----
Student ID: 14-2024-01
Violation Type: Late Check-in
Formatted Time: 2026-04-19 20:17:33
Note: Created by SYSTEM

Student ID: 14-2024-01
Violation Type: Late Check-in
Formatted Time: 2026-04-19 20:20:27
Note: Created by SYSTEM
```

```
----- Manager Dashboard -----
1 : Student Manage
2 : Manager Manage
3 : History & Violations
4 : Logout
Choice: 1
1 : Add Student
2 : Remove Student
3 : View Student Profiles
4 : Exit
3
----- Student Profiles -----
ID: 14-2024-01
Name: Jhustin Jane C. Alialy
Room: 210
Status: IN
-------------------
1 : Add Student
2 : Remove Student
3 : View Student Profiles
4 : Exit
```

```
----- Manager Dashboard -----
1 : Student Manage
2 : Manager Manage
3 : History & Violations
4 : Logout
Choice: 1
1 : Add Student
2 : Remove Student
3 : View Student Profiles
4 : Exit
3
----- Student Profiles -----
ID: 14-2024-01
Name: Jhustin Jane C. Alialy
Room: 210
Status: IN
-------------------
1 : Add Student
2 : Remove Student
3 : View Student Profiles
4 : Exit
```

```
----- Add Student -----
Enter student id: 14-2024-78
Enter student name: Lily Eleni R. Soliven
Enter student room: 413
Enter student pin: 0000
Student added.
1 : Add Student
2 : Remove Student
3 : View Student Profiles
4 : Exit
```

```
----- Student Profiles -----
ID: 14-2024-01
Name: Jhustin Jane C. Alialy
Room: 210
Status: IN
-------------------
ID: 14-2024-78
Name: Lily Eleni R. Soliven
Room: 413
Status: OUT
-------------------
1 : Add Student
2 : Remove Student
3 : View Student Profiles
4 : Exit
```

VII. Team Roles

Jhustin Alialy	**Ae-shin Barrete**	**Jemdrich Galido**	**Liam Gonzales**	**Jakob Pabico**
PPT, Assisted in coding, Problem Statement, Overview, Objectives	Flowchart for called API, API function, student access, and manager access	Assisted in coding, Problem Statement, Overview, Objectives	Methodology/ Technical Approach, Assisted in Output screenshots	Main code, READme file, GitHub repository

VIII. References

PEP 8 – Style guide for Python code | peps.python.org. (n.d.). Python Enhancement Proposals (PEPs). https://peps.python.org/pep-0008/